

Report_Los_Angeles_Data

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Introduction

In this report we present preliminary exploration of the LA dataset. We present different type of analysis we can do working on Sepulveda and Eldorado dataset.

Part 1 : Dataset description

1 - Surveillance effort

Starting in the late 80s, the dataset includes 161 traps of 6 types (BGSENT, CO2, GRVD, NJLT, USDS, OTHER?) accross 81 sites (trap location) in 2 areas (Sepulveda and Eldorado). Figure 1 show, until 2004 only 3 traps NJLT type were set up (2 in Sepulva and 1 in Eldorado). Then, we observe lot of variation in the number of trap set up.

- > All trap do not have the same number of monitor weeks.
- > Period of High monitoring after 2003.
- > Period of very intense monitoring between 2004 and 2010.

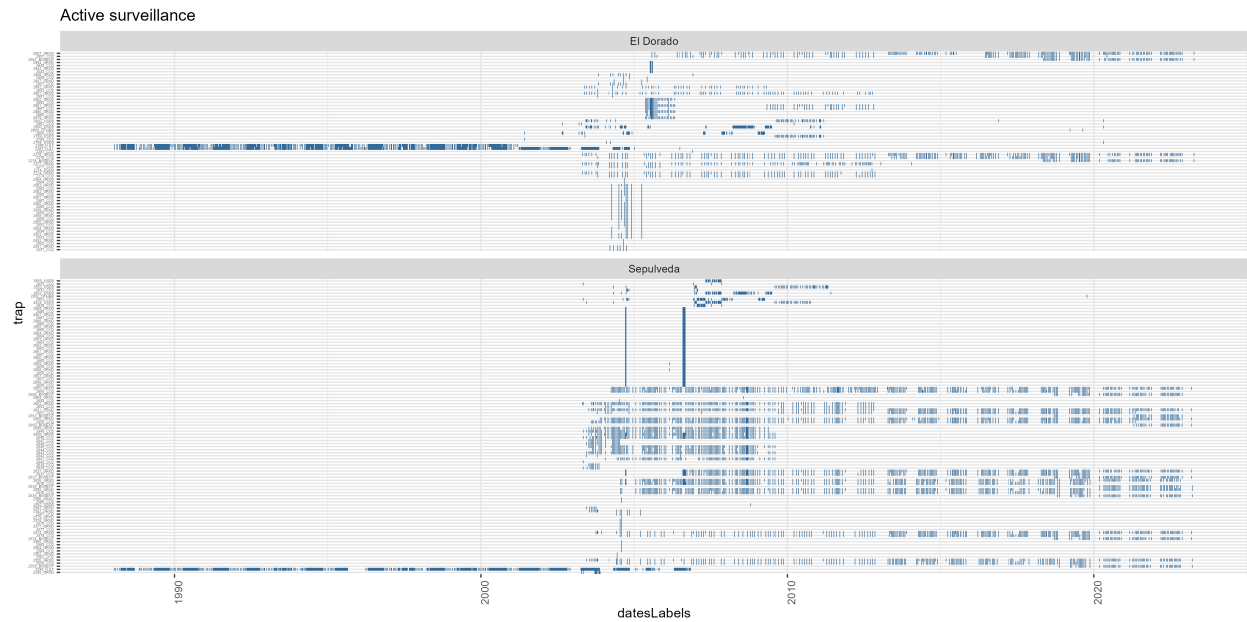


Figure 1: Active surveillance by trap (1 trap = 1 trap type & 1 location) per week in Eldorado (top) and Sepulveda (bottom). Each blue line represents a week during which the trap was active.

1.a. Surveillance effort over years

Surveillance efforts varies over years and areas. Higher effort of surveillance occurs around 2005 and then decrease. Early season monitoring decrease a lot after 2011 with almost no trap before midmay for last years. From 2005, the number of weeks monitor is higher in Sepulveda than in El Dorado.



Trap_ID = one location and one type of trap.

Figure 4a show (diapo 8) that the number of trap (one location, one type of trap at least set up once a year) each year vary. Some trap have been set up for years, other as been set up sporadically (Figure 4b).

1.b. Surveillance effort by trap type

Different type of trap are not used consistently. NJLT had been set up in only 3 locations, but was the only trap used until 2004 and so tthe trap with the highest effort (number of trap-nights). After 2004, CO2 and GRVD trap was the main trap used in many different location. USDS and BGSENT are used sporadically.

2 - Mosquito Diversity

19 different species has been identified with *Cx. quiquefasciatus* represent 93% of the total sample mosquitoes.

3 - Type of trap impact on specie

Trap performance to capture mosquitoes vary depending on species. Concerning the *Cx. quiquefasciatus* (except other) only NJLT capture less *Cx. quiquefasciatus* than average. *Is NJLT trap less efficient to capture Cx. quiquefasciatus?* According to the figure it seems that we get the general trend with only CO2 and NJLT trap (except for *Ae. notoscriptus* capture only once). "Other" type trap only catch *Ae. aegypti* is it Human-landing catch ?

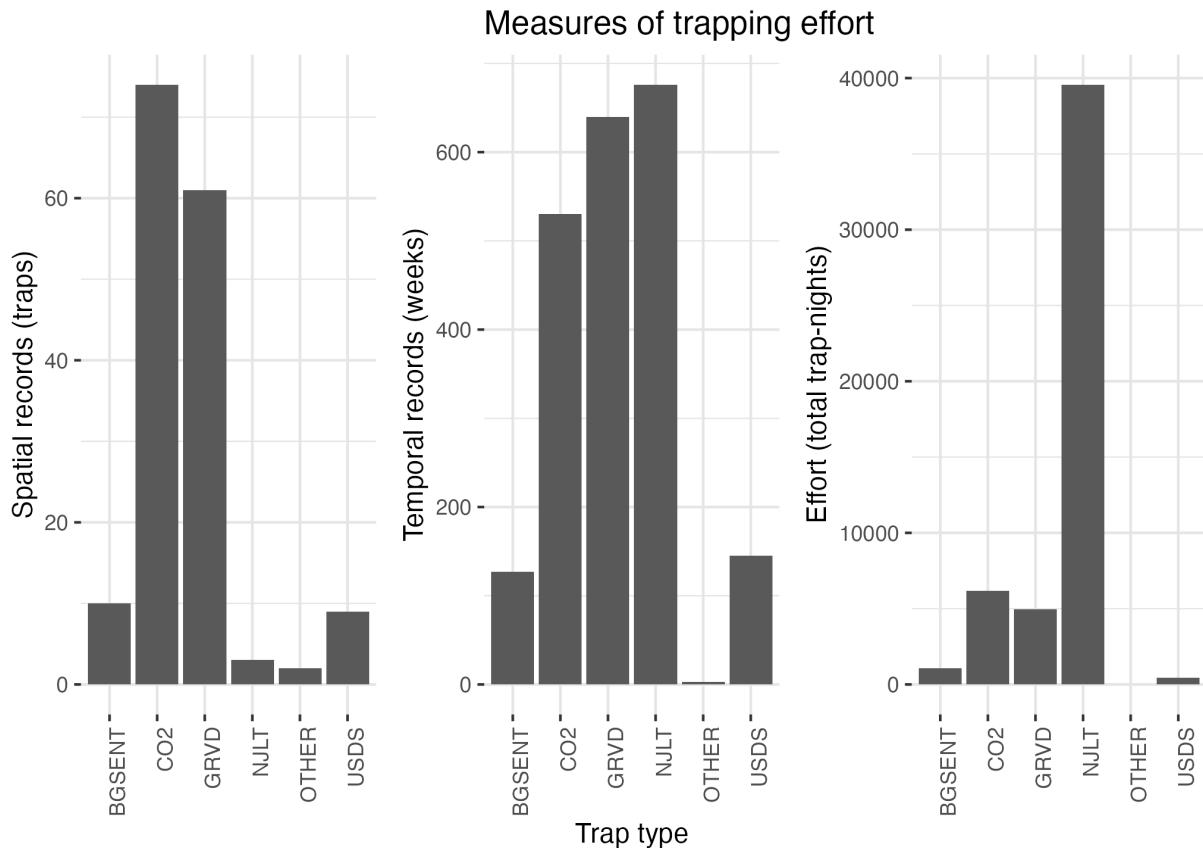


Figure 2: Trapping effort by type of Trap with number of location by trap type (left), number of weeks recorded by trap type (middle) and the total number of trap nights by trap type (right)

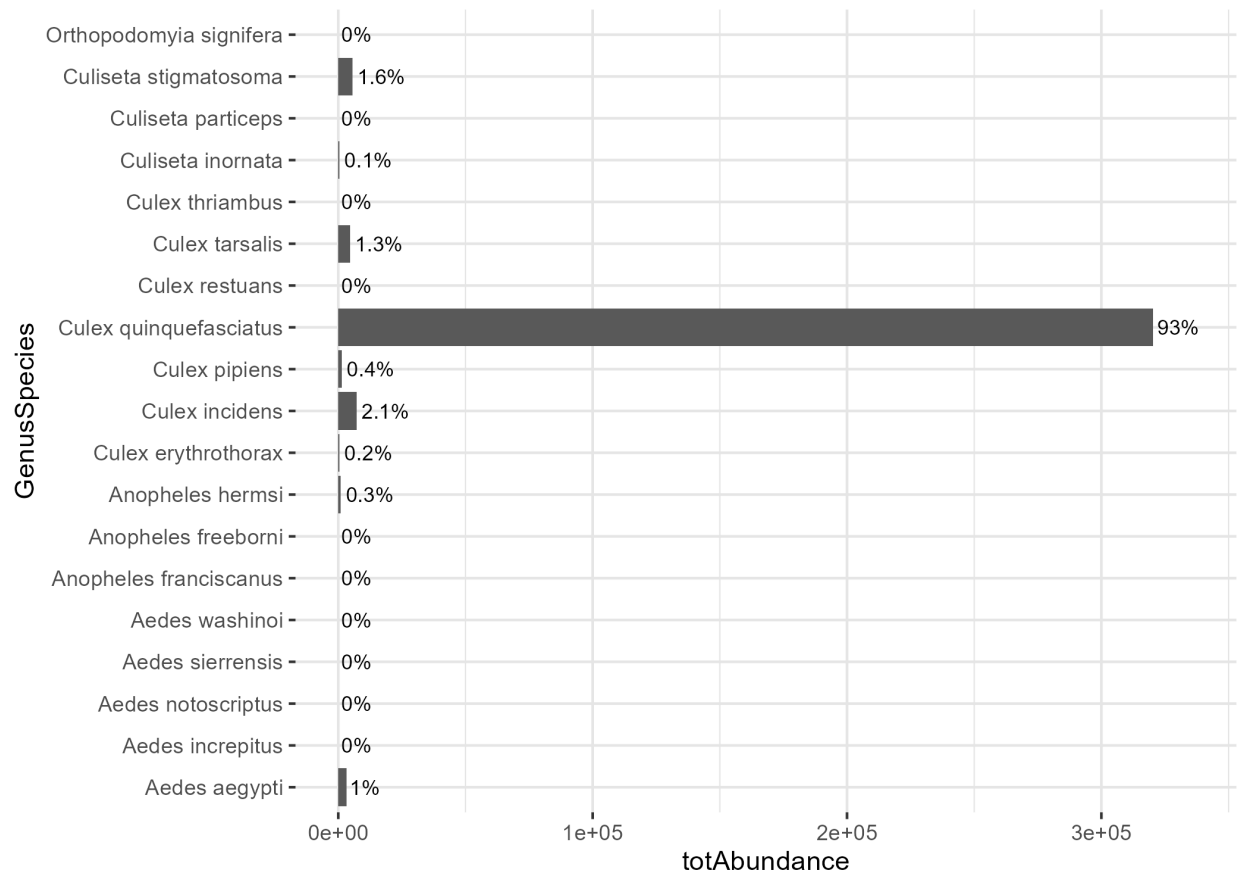
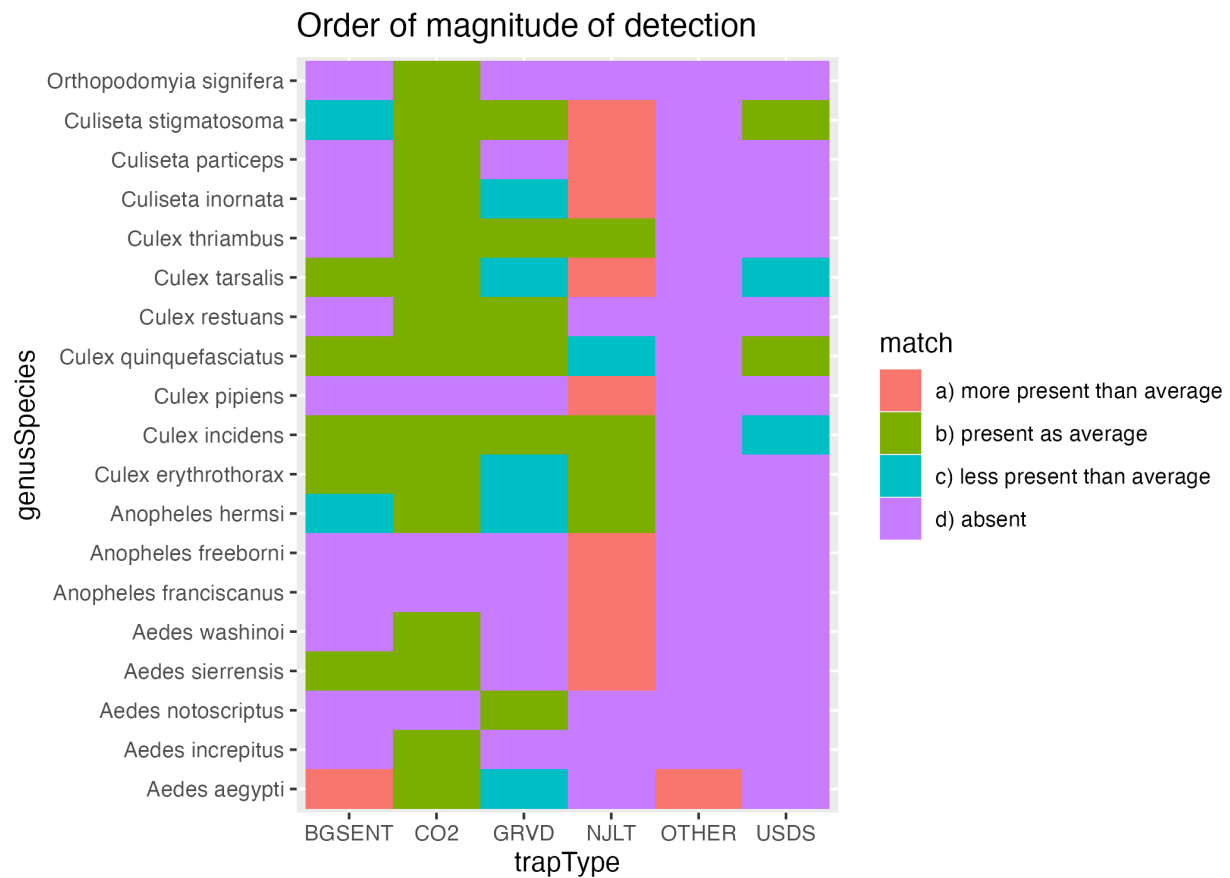


Figure 3: Trapping effort by type of Trap with number of location by trap type (left), number of weeks recorded by trap type (middle) and the total number of trap nights by trap type (right)

Figure 6 show that long old series NJLT trap do not capture a lot of *Cx. quinquefasciatus*.



Part 2 : Surveillance efficacy (minimal trapping effort, sentinel trap ...)